1. Figure 9

Code:
https://github.com/gphehub/pnn2204/additional_experiments_mar2026/figure9/fig9_saliency_map.py

Data:
https://github.com/gphehub/pnn2204/additional_experiments_mar2026/figure9/biases&weights_optimized.pickle

Copied from:
"experimental_data\ex1\ex1_1_[784,48,35,10]+[784,50,10]_sigmoid_60,100\Trial_1"
which is inside the file https://github.com/gphehub/pnn2204/experimental_data.zip

Can be replaced by biases&weights_optimized.pickle from other trials of the sigmoid function (The code fig9_saliency_map.py is designed for the network using the sigmoid function. Therefore it will not generate the correct saliency map for the networks using the ReLU and Tanh functions without modification).

Instruction for running the code:
(1) Preparation:
Put the above fig9_saliency_map.py, biases&weights_optimized.pickle, and mnist_loader.py in your working directory;
Put the MNIST dataset file mnist.pkl.gz in the /data subdirectory;
(mnist_loader.py and mnist_loader.py are not included here. They are vailable at:
https://github.com/mnielsen/neural-networks-and-deep-learning/archive/master.zip)

(2) Running the code:
Run fig9_saliency_map.py.
It will generate he_fig9.png and disagreed_results.csv in the working directory.

2. Table 3

Data:
The data for trials 10~23 is in the directory:
https://github.com/gphehub/pnn2204/additional_experiments_mar2026/table3/data

The data for trials 1~9 can be obtained as follows:
Download and unzip https://github.com/gphehub/pnn2204/experimental_data.zip
The data for the sigmoid function is in the unzipped directory:
experimental_data\ex1\ex1_4_[784,48,35,10]+[784,48,35,10]_sigmoid_0,100
The data for the ReLU and Tanh functions is in the unzipped directory:
experimental_data\ex4

Code:
The codes with the parameters already set for computing the data for Table 3 are in the directory:
https://github.com/gphehub/pnn2204/additional_experiments_mar2026/table3/code

Instruction for running the code:
(1) Preparation:
Put the above codes, all the codes in https://github.com/gphehub/pnn2204/code, and mnist_loader.py in your working directory;
Put the MNIST dataset file mnist.pkl.gz in the /data subdirectory;
(mnist_loader.py and mnist_loader.py are not included here. They are vailable at:
https://github.com/mnielsen/neural-networks-and-deep-learning/archive/master.zip)

(2) To compute new data for Table 3:
Run any of the following program:
main2204_2Para6S-Table3.py (for sigmoid)
main2204_2Para6R-Table3.py (for ReLU)
main2204_2Para6T-Table3.py (for Tanh)

They will create the following output files in the working directory:
biases&weights_initialized.pickle
biases&weights_optimized.pickle
classification_accuracy.csv
disagreed_results.csv
network2Para6S.pkl (or network2Para6R.pkl, or network2Para6T.pkl)

Open disagreed_results.csv, and use the data "Number of results when both subnetworks are wrong but the combined network is right" and "Para" as n_{IV} and α, respectively, for Table 3. The r_{IV} value in Table 3 is calculated as $(n_{IV}/10000)/\alpha$.

(3) To verify the existing data in Table 3:

Due to the randomness in the shuffling of the training data, loading biases&weights_initialized.pickle into the network and re-run the training process generally will *not* result in exactly the same existing data in Table 3.

Instead, to reproduce the existing data in Table 3, we should load biases&weights_optimized.pickle into the network using main2204_2Para_load_6R.py for ReLU (or main2204_2Para_load_6S.py for sigmoid, or main2204_2Para_load_6T.py for Tanh), and compute the classification result directly without re-training the network.

3. Table B.1

Data:
The data for Table B.1 is in the directory:
https://github.com/gphehub/pnn2204/additional_experiments_mar2026/tableB1/data

Code:
The codes for computing the data for Table B.1 are in the directory:
https://github.com/gphehub/pnn2204/additional_experiments_mar2026/tableB1/code

Instruction for running the code:
(1) Preparation:
Put the above codes in your working directory;
Put the CIFAR-10 dataset file cifar-10-python.tar.gz in the /data subdirectory;
(cifar-10-python.tar.gz is not included here. It is vailable at:
http://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz)

(2) To compute new data for Table B.1:
Run any of the following program:
PCNN2603-Sigmoid-CIFAR.py (for sigmoid)
PCNN2603-ReLU-CIFAR.py (for ReLU)
PCNN2603-Tanh-CIFAR.py (for Tanh)

They will create the following output files in the /data subdirectory:
biases&weights_initialized.pickle
biases&weights_optimized.pickle
classification_accuracy.csv
disagreed_results.csv

Open disagreed_results.csv, and use the data "Ratio of Type IV results to correct results" and "Accuracy on the validation data" as r_{IV} and α, respectively, for Table B.1.

(3) To verify the existing data in Table B.1:
Due to the randomness in the shuffling of the training data, loading biases&weights_initialized.pickle into the network and re-run the training process generally will *not* result in exactly the same existing data in Table B.1.
Instead, to reproduce the existing data in Table B.1, we should load biases&weights_optimized.pickle into the network using any of the above PY program, and compute the classification result directly without re-training the network.
The command for loading biases&weights_optimized.pickle:

```
with open('biases&weights_optimized.pickle', 'rb') as f:
    init_params = pickle.load(f)
```